

Early-stage research / prototype

<https://github.com/RandyTunru/episodic-latent-reasoning>

Chain-of-Thoughts Compression Through Episodic Latent Reasoning

Andi Analta Dwiyanto Palalangan Tunru

tunrurandy@gmail.com

<https://www.linkedin.com/in/analtadwiyanto/>



Premise:

Current autoregressive CoT scales quadratically, creating an inference bottleneck. This project proposes decoupling reasoning into a compressed, episodic latent space using the JEPA architecture to achieve sub-quadratic inference without losing logical depth.

Chain-of-Thought (CoT) prompting enables Large Language Models to execute complex reasoning by generating intermediate logical steps.

Human Prompt

A sailor's distance from a lighthouse is 500 meters. If the sailor is sailing directly towards the lighthouse at a constant speed of 10 m/s, how far will the sailor be from the lighthouse after 200 seconds?

Generated Response

Thinking
Block

<think>

First, I need to determine the distance the sailor covers by multiplying the speed by the time traveled. The speed is 10 meters per second, and the time is 200 seconds, so the distance covered is $10 * 200 = 2000$ meters.

Next, I'll subtract the distance covered from the initial distance to find the remaining distance from the lighthouse. The initial distance was 500 meters, so $500 - 2000 = -1500$ meters. Since distance cannot be negative, this indicates that the sailor has passed the lighthouse.

Therefore, the sailor is 1500 meters beyond the lighthouse.

</think>

Answer

To determine how far the sailor will be from the lighthouse after 200 seconds....

By explicitly generating these intermediate “thinking blocks”, the model creates a traceable context to condition its final response.

This autoregressive generation introduces severe bottlenecks. Thinking blocks routinely consume thousands of tokens, as each token would require $O(N)$ compute and $O(N^2)$ total for the full sequence.

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Furthermore, this approach couples abstract logical reasoning with linguistic syntax. [9][10]

Reasoning ability is commonly defined as its capability to traverse through multiple steps or hops of reasoning

<think>

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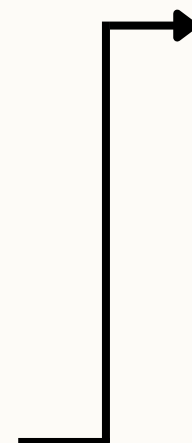
Therefore, the sailor is 1500 meters beyond the lighthouse.

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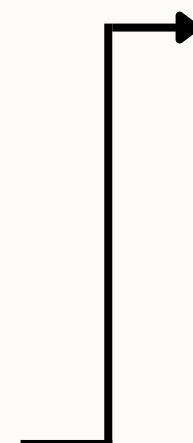
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Next, I'll subtract the distance covered from the initial distance to find the remaining distance from the lighthouse.



The initial distance was 500 meters, so $500 - 2000 = -1500$ meters



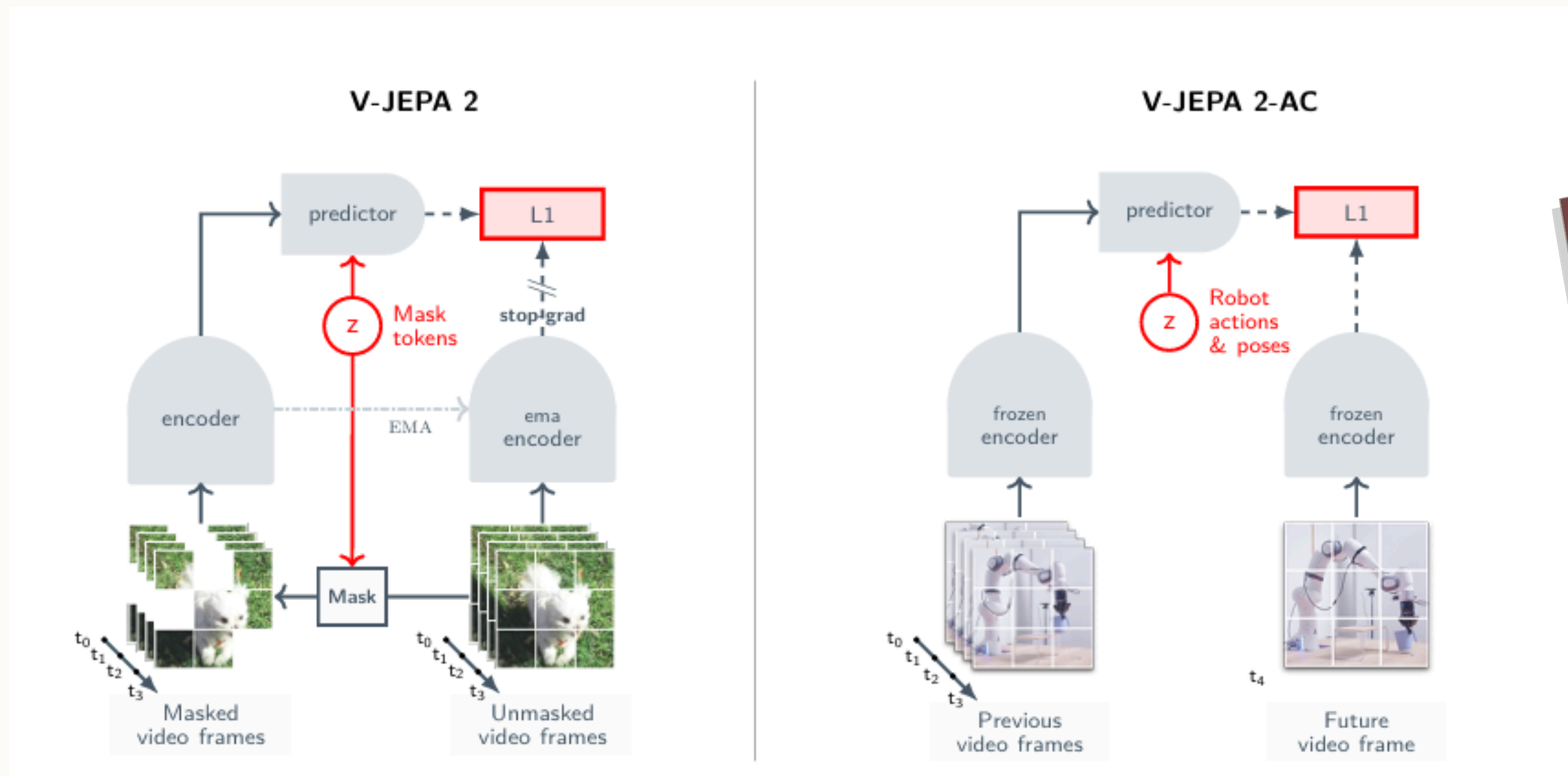
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Therefore, the sailor is 1500 meters beyond the lighthouse.

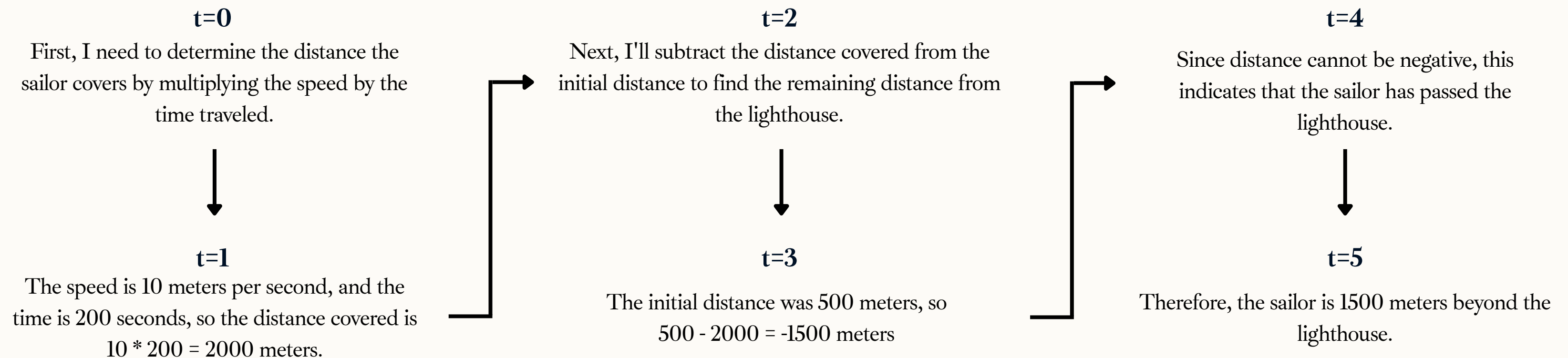
Joint Embedding Predictive Architecture (JEPA) has recently showed interesting results being able to predict directly in latent space rather than pixel/token space, avoiding the wasteful step of reconstructing raw inputs.

A variant V-JEPA 2 [5] had even shown capability of predicting future video frame given previous video frames as context and conditioned on the action happening



By first training an encoder to learn high-quality representations, it can then be used with a separate predictor model that takes the context from the trained encoder and a given action conditioning (AC) to predict the output target latent space of a future frame that is also processed using the same trained encoder model

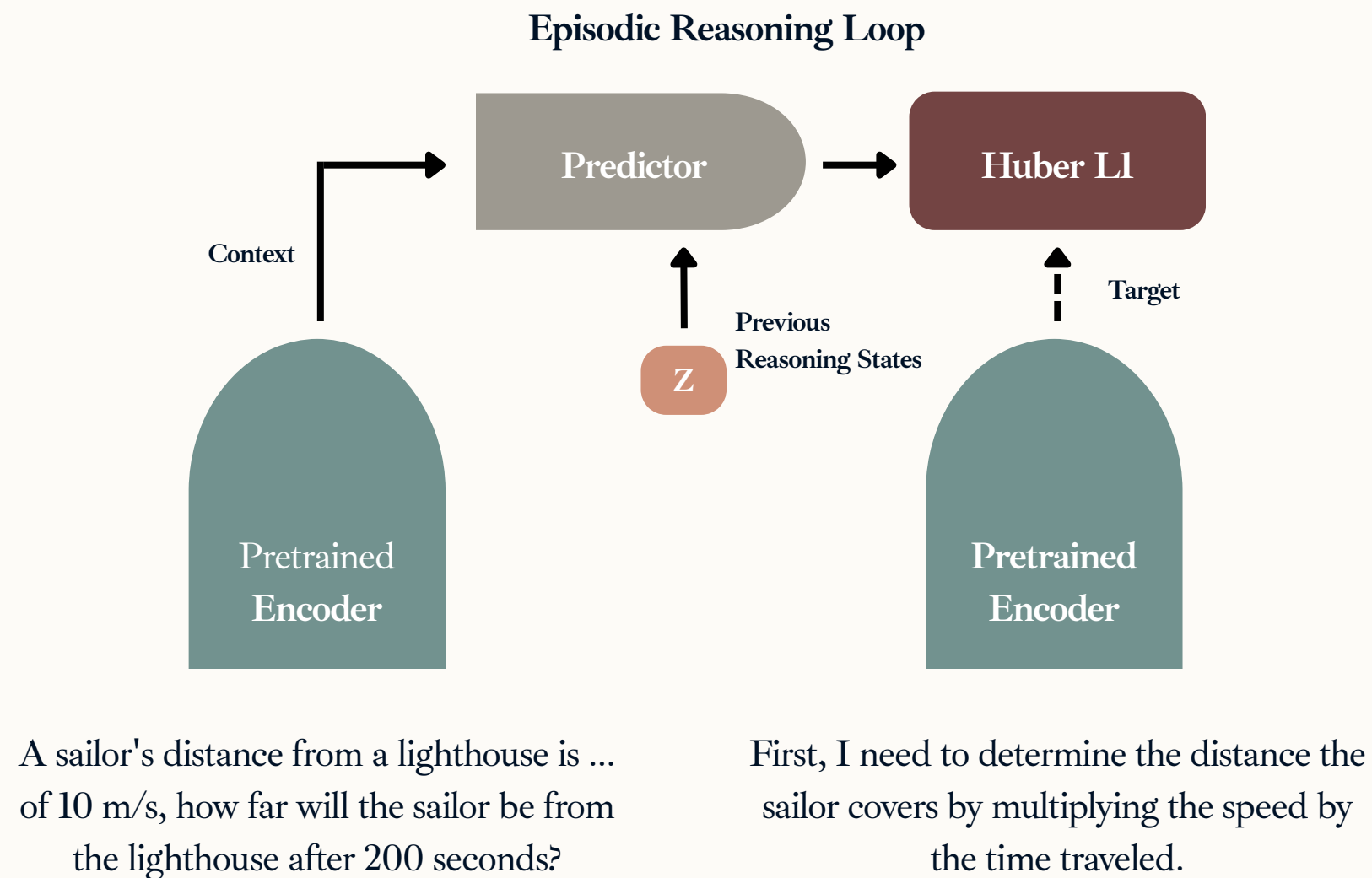
Sequential reasoning steps can be modeled as episodic state transitions, similar to temporal frames in a video sequence.



Using the same concepts introduced in V-JEPA 2, it is interesting to explore if we can encode these reasoning steps as discrete state-space transitions over time, where the JEPA predictor learns the transition dynamics of logic, much like it learns the physical dynamics of video frames.

Proposed Approach: Dual module, separate models reasoning and language

1. JEPA-based reasoning module



A frozen, pretrained encoder provides the static semantic context, while an autoregressive Predictor iteratively forecasts sequential latent reasoning states Z . A jointly trained, lightweight router dynamically evaluates these states to trigger a halting mechanism upon logical convergence similar to LoopLM [2][6] approaches.

The predictor is an autoregressive process which uses the output of the target encoder as its continuous ground truth and relies on a learnable <start> token to initiate the recurrent prediction loop.

Proposed Approach: Dual module, separate models reasoning and language

2. Language Model

A sailor's distance from a lighthouse is 500 meters. If the sailor is sailing directly towards the lighthouse at a constant speed of 10 m/s, how far will the sailor be from the lighthouse after 200 seconds?

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To determine how far the sailor will be from the lighthouse after 200 seconds....



A sailor's distance from a lighthouse is 500 meters. If the sailor is sailing directly towards the lighthouse at a constant speed of 10 m/s, how far will the sailor be from the lighthouse after 200 seconds?

<think>

<reasoning_state_1> <reasoning_state_2> <reasoning_state_3>....

<reasoning_state_n>

</think>

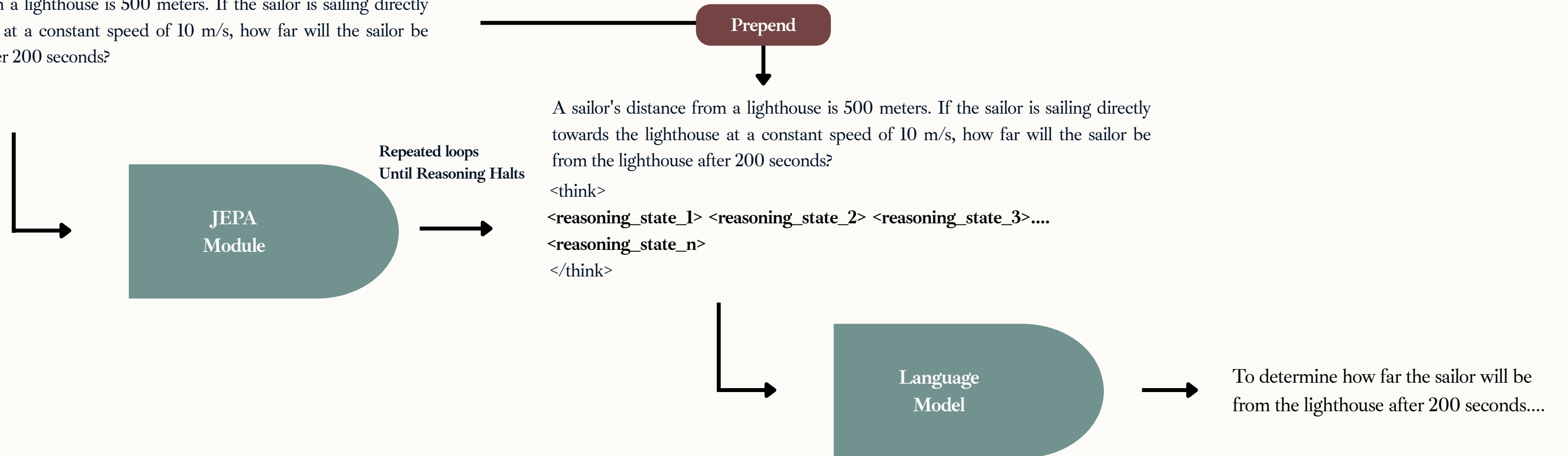
To determine how far the sailor will be from the lighthouse after 200 seconds....

Building on previous research [3][4][10], the predicted latent reasoning states are injected directly into the LLM's context window as continuous embeddings. The language model is then fine-tuned to natively process standard textual tokens alongside these dense, latent reasoning representations.

Proposed Approach: Dual module, separate models reasoning and language

3. Inference

A sailor's distance from a lighthouse is 500 meters. If the sailor is sailing directly towards the lighthouse at a constant speed of 10 m/s, how far will the sailor be from the lighthouse after 200 seconds?



In a single pass, our context must first be passed onto the reasoning JEPA module which would output the reasoning state tokens. These states are then concatenated with the original context and passed to the language model to generate the final response

This mirrors the soft-prompting techniques used in Vision-Language Models (VLMs), allowing the LLM to cross-attend to dense reasoning representations

Potential Gains:

1. Compressing verbose CoT into a sequence of dense latent tokens drastically reduces KV-cache pressure and accelerates generation speed.
2. Decoupling logical planning from syntactic token generation, the architecture prevents minor grammatical, or vocabulary slips from derailing the reasoning trajectory
3. Enables gradient-based optimization of reasoning trajectories. Unlike discrete token search, which requires reinforcement learning or beam search

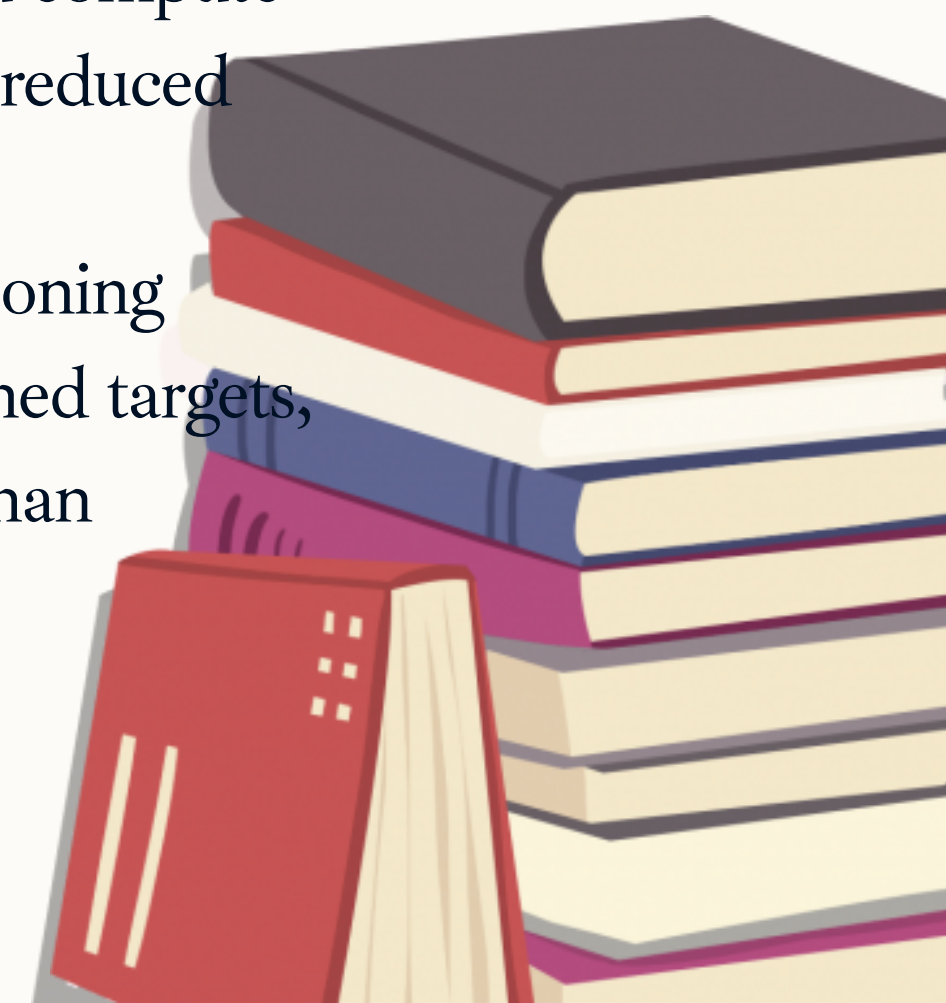
Why now?

1. The scaling cost of textual CoT is unsustainable. A growing body of work (Coconut, Token Assorted, CoLaR, ImgCoT, 2024–2026) are exploring continuous-space alternatives
2. Latent prediction for structured dynamics are no longer speculative. V-JEPA 2 demonstrated that an autoregressive predictor can learn transition dynamics between compact latent states when conditioned on a frozen encoder's context. Reasoning steps are a natural candidate for the same treatment.
3. Training data is no longer the bottleneck. High-quality explicit reasoning traces with clear step structure (DeepSeek-R1, Magpie-Reasoning) are now plentiful, open-source, and parseable into the step-level format our approach requires.
4. PEARL [12] showed predictive embedding alignment beating reconstruction-based latent training in multimodal tool-use; its authors flag test-time reasoning over latent states as the open next step. This project takes that step



Acknowledged Challenges & Opportunities:

1. **Interpretability is reduced.** Compressing reasoning into continuous latent states means we lose the ability to read and audit the model's logic step-by-step. We trade transparency for efficiency
2. **Information bottleneck.** The compressed latent states must preserve enough detail (numbers, entities, logical relationships) for the downstream language model to produce a correct answer. If the predictor's dimensionality is too aggressive, critical information is lost silently.
3. **Parameter overhead vs. KV-cache savings.** The JEPA module adds parameters and compute that wouldn't exist in a standard CoT pipeline. The net win depends on whether the reduced KV-cache pressure in the primary LLM outweighs the cost of running the predictor
4. **Step segmentation is a source of bias.** Breaking a continuous CoT into discrete reasoning steps requires a heuristic (e.g., paragraph splits). Poor segmentation produces misaligned targets, which can bias the predictor toward learning artifacts of the splitting strategy rather than genuine logical transitions.



Design Choices and Experiments made

1. **Shared encoder design.** The context branch reuses the target LLM's own frozen backbone no separate encoder copy, and its KV cache is shared with the LLM's prefill. Compared against a dedicated embedding model for context, both perform comparably and beat mean-transition and identity-transition baselines. (addressing challenge #3)
2. **Canonicalized chain-of-thought traces.** Real CoT interleaves genuine steps with meta-thoughts ("Okay, let me try to figure this out", "Hmm, maybe I should check this again"), self-corrections, and restatements. We rewrite each trace offline into clean, numbered logical steps before encoding so the predictor learns logic transitions rather than narration artifacts, and the position-derived halt labels stay clean. (addressing challenge #4)



Current Progress:

Collecting chain-of-thoughts datasets and synthetically modifying thinking blocks into reasoning steps

Model and Experiment Designs:
Model architecture ✓
Data pipeline ✓
Training loop ✓

Current Implementations in <https://github.com/RandyTunru/episodic-latent-reasoning>

Training and evaluating model. Analysis of performance and comparative studies

Currently Here

Papers Consulted:

1. Hao et al. - Training Large Language Models to Reason in a Continuous Latent Space
2. Geiping et al. - Scaling up Test-Time Compute with Latent Reasoning: A Recurrent Depth Approach
3. Su et al. - Token Assorted: Mixing Latent and Text Tokens for Improved Language Model Reasoning
4. Tan et al. - Think Silently, Think Fast: Dynamic Latent Compression of LLM Reasoning Chains
5. Assran et al. - V-JEPA 2: Self-Supervised Video Models Enable Understanding, Prediction and Planning
6. Bae et al. - Mixture-of-Recursions: Learning Dynamic Recursive Depths for Adaptive Token-Level Computation
7. Bui et al. - Speaking in Words, Thinking in Logic: A Dual-Process Framework in QA Systems
8. Huang et al. - LLM-JEPA: Large Language Models Meet Joint Embedding Predictive Architectures
9. Liu et al. - JEPA-Reasoner: Decoupling Latent Reasoning from Token Generation
10. Chen et al. - ImgCoT: Compressing Long Chain of Thought into Compact Visual Tokens for Efficient Reasoning of Large Language Model
11. Amos et al. - Latent Reasoning with Supervised Thinking States
12. Adhikari et al. - Multimodal Latent Reasoning via Predictive Embeddings

Thank You!

Comments and criticism are highly appreciated

tunrurandy@gmail.com